BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: PR IX (01) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Physics Paper–IX (01) (21)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Force per unit area is called.....

(A) Pressure
(B) Strain
(C) Power
(D) Work

Q2. Work will be zero when the angle between force and displacement is.....

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q3. The temperature at which a body is not radiating heat is.....

(A) 0°C
(B) 0°F
(C) 0 K
(D) All these

Q4. Conventionally anti clock wise torque is taken as:

(A) Negative
(B) Positive
(C) Parallel
(D) Zero

Q5. The amount of heat required to raise the temperature of 1 kg of water by 1 K is.....

(A) [illegible in source scan]
(B) 400 J
(C) 310 J
(D) 4190 J

Q6. Young modulus is measured in units of.....

(A) Kg m^{-3} (B) Pa

(C) N m^{-2} (D) None of these

Q7. The average speed of a bus is 20 ms^{-1}, how far can it travel in 10s?

(A) 100 m (B) 200 m

(C) 150 mm (D) 250 m

Q8. Which of the given is base quantity?

(A) Volume (B) Density

(C) Length (D) Area

Q9. Turning effect produced in a body is called.....

(A) Force (B) Torque

(C) Power (D) None of these

Q10. The product of force and displacement gives.....

(A) Work (B) Momentum

(C) Power (D) None of these

Q11. The S.I unit of power is.....

(A) Watt (B) Joule

(C) Kelvin (D) Newton

Q12. The resultant force of a couple is:

(A) Negative (B) Positive

(C) Maximum (D) Zero

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. How technology is shaped by Physics?

2. Can an object have zero acceleration and non-zero velocity at the same time? Give an example.

3. Why does dust fly off, when a hanging carpet is beaten with a stick?

4. What is couple? Explain it with an example.

5. Differentiate between heat and temperature.

6. How woolen sweaters keep us warmer in winter?

7. Explain anomalous expansion of water.

8. Explain different types of motion by giving one example of each.

9. A 60 g bullet is fired from a gun with 3150 J of K.E. Find its velocity.

10. Moon is attracted by the earth, why it does not fall on earth.

11. Why a tile floor feels colder to bare feet than carpeted floor?

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] State and explain international system of units.

(b) [3] What is the acceleration produce by a force of 12 N exerted on an object of Mass 3 kg?

QIV.

(a) [4] Prove that vf = vi + at using graphical method.

(b) [3] To open a door force of 15 N is applied at 30° to the horizontal, find the horizontal and vertical components of force.

QV.

(a) [4] State and explain potential energy. Prove that EGP.E = mgh

(b) [3] How much work can a 2.0 – hp motor do in 1.0 hour?

QVI.

(a) [4] State and explain law of universal gravitation.

(b) [3] What is the specific heat of a metal substance if 135 KJ of heat is needed to raise 4.1 kg of the metal from 18.0°C to 37.2°C?